Project 1 – Small Office Network

Nov 3, 2026 Usama Kakembo · Network Engineering portfolio · Github.com/usamakaks

Overview

I designed and built a secure, segmented network for a 30-person accounting firm, from requirements to a verified build in Cisco Packet Tracer. It has 5 VLANs, router-on-a-stick inter-VLAN routing, DHCP, PAT to a single public IP, guest isolation, SSH-only management, port security and DHCP snooping.

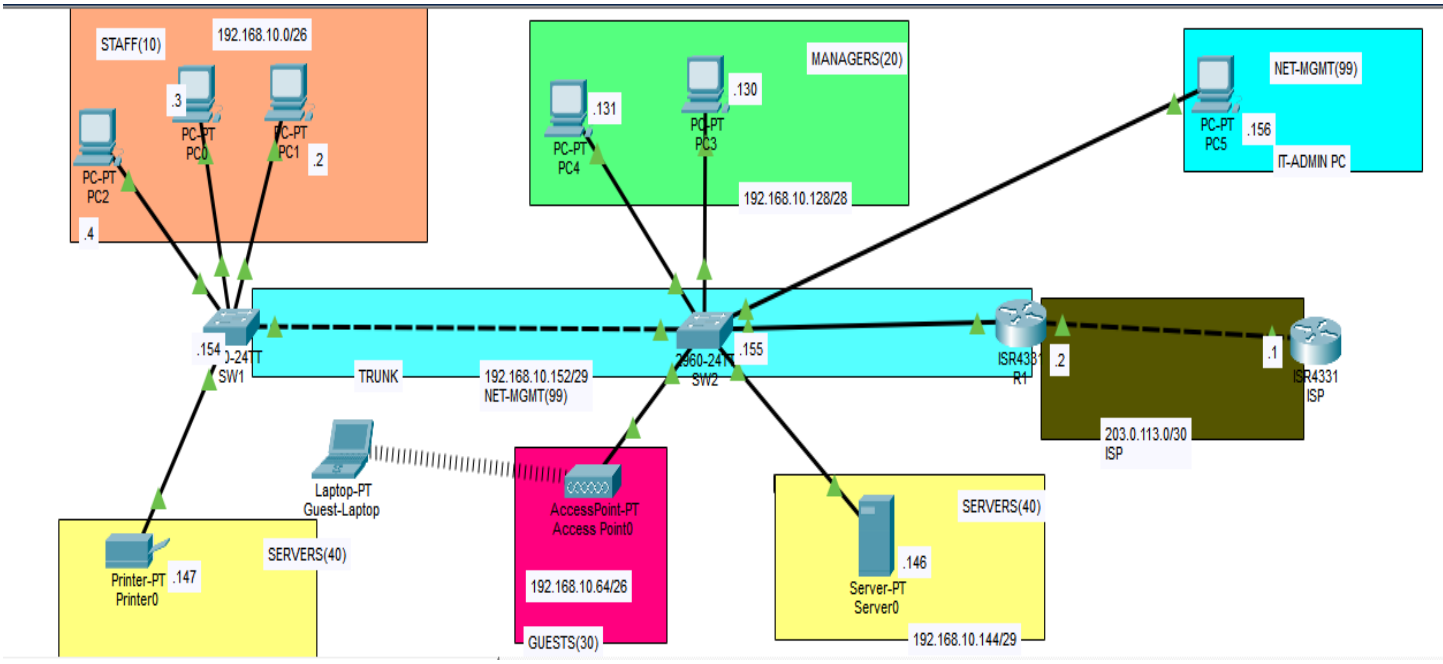
- **Tools:** Cisco Packet Tracer, Cisco IOS CLI
- **Devices:** 1 ISR4331 router (R1), 2 Catalyst 2960 switches (SW1, SW2), 1 ISP router, PCs, a server, a printer, a wireless access point
- **Address space:** 192.168.10.0/24 internal (RFC 1918), 203.0.113.2/30 public

Requirements

The client is a small accounting firm on one floor, expecting about 50% growth in two years.

Group	Devices today	Needs
Staff	25 wired PCs	Internet + file server
Managers	5 wired PCs	Same as staff, on a separate network
Servers	1 file server, 1 printer	Fixed IP addresses
Guests	Up to 20 Wi-Fi devices	Internet only, nothing internal
IT	Network devices	Secure remote management
Internet	1 public IP from the ISP	203.0.113.2/30, ISP gateway .1

Topology



Project 1 logical topology · 4 network devices, 6 endpoint groups

R1 is the only router: it is the gateway for every VLAN, hands out DHCP and translates user traffic to the ISP.

Device	Model	Role
R1	ISR4331	Gateway for all VLANs, DHCP, PAT, ACLs
SW1	Catalyst 2960	Access switch: staff, printer
SW2	Catalyst 2960	Access switch: managers, server, guest AP, IT-ADMIN; uplink to R1
ISP	Router	Simulated provider; loopback 8.8.8.8 stands in for the internet
Access point	AccessPoint-PT	SSID OFFICE-GUEST, WPA2-PSK, bridged into VLAN 30

VLAN table

Seven VLANs keep each group in its own broadcast domain; VLAN 1 has no ports. Port numbers match the Packet Tracer build

VLAN	Name	Purpose	SW1 ports	SW2 ports
10	STAFF	Staff PCs	Fa0/1–3	—
20	MANAGERS	Manager PCs	—	Fa0/1–2
30	GUESTS	Guest Wi-Fi via access point	—	Fa0/5
40	SERVERS	File server, printer	Fa0/4 (printer)	Fa0/4 (server)
99	NET-MGMT	Switch/router management, IT-ADMIN PC	SVI .154	SVI .155, Fa0/7
998	UNUSED	Parking VLAN for shut-down ports	Fa0/6–24, Gi0/1–2	Fa0/8–24, Gi0/1–2
999	NATIVE	Native VLAN on trunks, carries no users	trunk	trunk

Trunks

Link	Ports	Allowed VLANs	Native
SW2 ↔ R1	SW2 Fa0/3 ↔ R1 G0/0/0	10, 20, 30, 40, 99	999
SW1 ↔ SW2	SW1 Fa0/5 ↔ SW2 Fa0/6	10, 40, 99	999

IP addressing plan

Each subnet is sized for today's users plus 50% growth plus the gateway, allocated largest first (VLSM) from 192.168.10.0/24. Gateways use the first usable address; 192.168.10.160–.255 is kept free for future VLANs.

VLAN	Need	Subnet	Mask	Gateway	Usable	Broadcast
10 STAFF	38	192.168.10.0/26	255.255.255.192	.1	.2 – .62	.63
30 GUESTS	30	192.168.10.64/26	255.255.255.192	.65	.66 – .126	.127
20 MANAGERS	8	192.168.10.128/28	255.255.255.240	.129	.130 – .142	.143
40 SERVERS	3	192.168.10.144/29	255.255.255.248	.145	.146 – .150	.151
99 NET-MGMT	4	192.168.10.152/29	255.255.255.248	.153	.154 – .158	.159
WAN	2	203.0.113.0/30	255.255.255.252	ISP .1	R1 .2	.3

Static assignments

Device	Address
R1 subinterfaces	.1, .65, .129, .145, .153
SW1 / SW2 (VLAN 99 SVI)	.154 / .155
IT-ADMIN PC	.156
File server	.146
Printer	.147
R1 G0/0/1 (WAN)	203.0.113.2
ISP loopback (simulated internet)	8.8.8.8

DHCP pools on R1

Pool	Excluded	Default router	DNS
STAFF	.1 – .10	192.168.10.1	8.8.8.8
MANAGERS	.129 – .131	192.168.10.129	8.8.8.8
GUESTS	.65 – .70	192.168.10.65	8.8.8.8

Servers and NET-MGMT have no pool: infrastructure stays static.

Build phases

The network was built bottom-up in six phases, verifying each before starting the next.

1. **Layer 2:** VLANs, access ports, trunks with native VLAN 999 and pruned allowed lists, unused ports shut down in VLAN 998. Verified with `show vlan brief`, `show interfaces trunk`.
2. **Inter-VLAN routing:** R1 subinterfaces with `encapsulation dot1Q`, switch management SVIs with `ip default-gateway`. Verified with pings across VLANs.
3. **DHCP + guest Wi-Fi:** three pools on R1 with excluded ranges; WPA2-PSK SSID on the access point. Verified with `show ip dhcp binding`.
4. **Internet:** default route to the ISP and PAT (NAT overload) for user VLANs only. Verified with `show ip nat translations`.
5. **Security:** guest isolation ACL, SSH-only management, port security, DHCP snooping. Verified with ACL hit counters, `show port-security`, `show ip dhcp snooping`.
6. **Verification and documentation:** full test checklist and this document.

Security policy

Access follows least privilege: each group reaches only what it needs.

Source	Internal network	Internet	Device management
Staff	Yes	Yes	No
Managers	Yes	Yes	No
Guests	No	Yes	No
Servers / printer	Yes	No	No
Network devices	Yes	No	—
IT-ADMIN PC	Yes	No	SSH only

Controls that enforce it

Control	Where	What it does
ACL `GUEST-IN` (extended)	R1 G0/0/0.30, inbound	Permits DHCP, denies 192.168.10.0/24, permits the rest
ACL `NAT-USERS` (standard)	R1 PAT rule	Only STAFF, GUESTS, MANAGERS are translated to 203.0.113.2
ACL `MGMT-ACCESS` (standard)	VTY lines on R1, SW1, SW2 (`access-class`)	Only 192.168.10.156 may log in; others denied and logged
SSH v2, local users	R1, SW1, SW2	`transport input ssh`, `login local`, RSA 2048, no Telnet
Port security	User, server, printer, AP ports	Sticky MACs; `restrict` on user/AP ports, `shutdown` on server, printer, IT-ADMIN; AP max 64
DHCP snooping	SW1, SW2, VLANs 10/20/30	Only uplinks toward R1 are trusted (SW1 Fa0/5, SW2 Fa0/3)
Unused ports	SW1, SW2	Shut down, access mode, VLAN 998
Trunk hardening	All trunks	Native VLAN 999, pruned allowed lists, `nonegotiate` on access ports

Design decisions

Decision	Why
Size subnets for +50% growth	Re-addressing later means changing every device, DHCP scope and ACL
Separate SERVERS and NET-MGMT VLANs	Servers and network devices can be protected with their own ACLs

Decision	Why
Two switches, not one	25 + 5 users plus server, printer, AP and uplink exceed 24 ports
Router-on-a-stick	One router interface serves five VLANs; simplest design at this size
Gateway = first usable address	Predictable for anyone on the team; leaves .2/.3 free for HSRP later
Prune trunks (SW1 carries 10, 40, 99 only)	No unnecessary broadcasts or VLAN paths; add with `allowed vlan add`
Native VLAN 999, no VLAN 1 ports	Prevents VLAN-hopping and keeps default VLAN empty
Parking VLAN 998 + shutdown	Two layers: a port re-enabled by mistake still lands in a dead VLAN
DHCP on R1	R1 already has an interface in every VLAN, so no relay is needed
PAT for user VLANs only	Least privilege: servers and network devices never need the web
Guest ACL inbound on the guest subinterface	Extended ACLs go closest to the source; drops traffic at the first hop
SSH from one admin PC	Encrypted logins, named users, no logins from user desks
`restrict` for users, `shutdown` for infrastructure	Users swap cables; a new MAC on the server port means something is wrong
No port security on trunks or sticky on AP	Trunks and the AP legitimately carry many changing MACs

Problems and fixes

Each problem was found from verification output, not by guessing.

Phase	Symptom	Cause	Fix
1	Only one trunk in `show interfaces trunk`; Fa0/3 in VLAN 1	SW2 port to R1 left as an access port	Configured Fa0/3 as a trunk, native 999, allowed 10,20,30,40,99
1	R1 missing from `show cdp neighbors`; Fa0/3 notconnect	Router interfaces are shut down by default	`no shutdown` on R1 G0/0/0
4	Named ACL commands rejected	Named ACL typed on one line, and `standard` added to the NAT command	Created the ACL, then entered `permit` lines inside it
4	Server still reached 8.8.8.8 after new NAT ACL	Old NAT rule (permitting everything) still active	Cleared translations, removed old rule and leftover ACLs 1–5
5a	Staff → guest ping appeared to work	Pinged the gateway .65 instead of the guest; ACL counters exposed it	Retested against the guest's real IP; ping failed as designed
5b	VTY ACL had no effect	`access-class` referenced an ACL that did not exist yet	Created `MGMT-ACCESS` in two steps; added `deny any log`
5c	Port security not enforced; trunks inside the range	Only `mac-address sticky` was set, on ranges that included trunks	Enabled `switchport port-security` on edge ports only
5c	Staff PCs would lose DHCP	DHCP snooping on SW1 with no trusted uplink	`ip dhcp snooping trust` on SW1 Fa0/5

Verification checklist

Run every test in one pass after the build; set Result as you go. A failure here means a later phase broke an earlier one.

Test	From	Command	Expected	Result
Ping own gateway	Each VLAN	`ping <gateway>`	Success	Success
Inter-VLAN	Staff → manager → server	`ping 192.168.10.132`, `.146`	Success	Success
Internet	Staff, manager, guest	`ping 8.8.8.8`	Success	Success
Internet blocked	Server	`ping 8.8.8.8`	Fails	Fails
Guest isolation	Guest laptop	`ping 192.168.10.146`, `.11`	Fails	Fails
DHCP	Staff, manager, guest	Renew in IP Configuration	Correct subnet + gateway	Correct
SSH allowed	IT-ADMIN	`ssh -l admin 192.168.10.153/.154/.155`	Login	Login
SSH blocked	Staff PC	`ssh -l admin 192.168.10.154`	Refused	Refused
Telnet blocked	IT-ADMIN	`telnet 192.168.10.154`	Refused	Refused
Port security	New laptop on a staff port	`ping 192.168.10.1`	Blocked, violation count rises	Blocked
DHCP snooping	SW1, SW2	`show ip dhcp snooping binding`	Leases listed per VLAN	Leases listed per VLAN

Configs and file

All configs and the .pkt file: github.com/usamakaks

Lessons learned

- Know what correct output looks like before running a `show` command.
- Adding config never removes old config; check `show running-config` after every change.
- Communication needs a path in both directions: ask whether the request or the reply failed.
- ACL hit counters are proof; a ping result alone can mislead.
- Named ACLs are a mode you enter, not a single line.
- Least privilege applies everywhere: trunks, NAT, management access, switch ports.

Next steps: Project 2 builds a two-building school campus: hierarchical design, multilayer switches with SVIs, EtherChannel, Rapid STP, HSRP and a wireless LAN controller.